

Tufte — one-page cheatsheet

Distilled from VDQI (1983/2001), Envisioning Information (1990), Visual Explanations (1997).

THE FIVE SUMMARY RULES (VDQI, P. 105)

1. Above all else show the data.
2. Maximize the data-ink ratio.
3. Erase non-data-ink.
4. Erase redundant data-ink.
5. Revise and edit.

GRAPHICAL EXCELLENCE (VDQI, P. 13)

- Show the data
- Induce the viewer to think about *substance*, not methodology or graphic design
- Avoid distorting what the data have to say
- Present many numbers in a small space
- Make large data sets coherent
- Encourage the eye to compare different pieces of data
- Reveal data at several levels of detail — overview to fine structure
- Serve a reasonably clear purpose
- Be closely integrated with statistical and verbal descriptions

GRAPHICAL INTEGRITY — LIE FACTOR

Lie factor = (effect shown in graphic) / (effect in data)

- Keep between 0.95 and 1.05
- Bar charts: y-axis at zero
- Line charts: label baseline if truncated
- Never use area or volume for a 1D quantity
- Avoid 3D perspective and dual axes

SMALLEST EFFECTIVE DIFFERENCE (VE, P. 73)

"Make all visual distinctions as subtle as possible, but still clear and effective."

- Gray for context, one accent color for the focal value
- If light gray reads, don't use dark gray
- If thin reads, don't use thick
- If 10pt reads, don't use 12pt

COLOR RULES

- Color labels, quantifies, decorates — never confuses
- For ordered data: sequential single-hue (light → dark)

THE CHART-SELECTION TABLE

DATA + GOAL	USE
Many items, compare	Sorted dot plot
One variable over time	Line, range frame
Two variables, relate	Scatter, no grid
Two snapshots, change	Slopegraph
Distribution	Strip plot
Part-to-whole, $n \leq 6$	Sorted bar or table
Cross-cut comparison	Small multiples
Geographic	Sequential single-hue
Number + trend	Sparkline + value
$n \leq 15$ exact values	Table

KILL LIST

KILL	REPLACE WITH
Pie chart	Sorted bar or table
3D anything	2D version
Dual y-axis	Two small charts
Rainbow scale	Single-hue sequential
Heavy gridlines	Light gray, or omit
Chart border	Range frame
Legend (≤ 4 series)	Direct labels
KPI tile alone	Number + sparkline + delta
Gradient bar fills	Flat fill
Donut, gauge, radar, word cloud	Sorted bar or table

PRE-PUBLISH CHECKLIST (30 SEC)

1. Sorted by what the reader cares about?
2. Lie factor 0.95-1.05?
3. Bar charts zero-anchored?
4. Truncated lines have baseline labeled?
5. Gridlines light gray or absent?
6. No frame box (range frame only)?
7. No redundant encoding?
8. One accent color, used purposefully?
9. Direct labels on data?
10. Numbers formatted (4.2M not 4,200,000)?
11. Could a small table do this job?
12. Could one sentence do this job?

- For diverging data: two-hue (e.g., blue-white-red), zero at white
- For categorical: maximum 4 colors, then group or use small multiples
- Accent color (red/orange) earns its place — never default
- Reds and greens together: add luminance difference for colorblind readers

The principles are universal — like mathematics — and not tied to a particular language or culture. — Tufte, VDQI, p. 11.